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Lubrication and cooling of equipment of an aircraft turbomachine

Abstract

An aircraft turbomachine includes a gas generator having an output shaft and a first lubricating circuit. The turbomachine further includes equipment coupled to the output shaft and including a rotor rotationally guided by at least one rolling bearing and a second lubricating circuit that is independent of the first lubricating circuit and is configured to lubricate the rolling bearing. The equipment further includes a system for cooling the rolling bearing, the cooling system being configured to circulate oil in the region of at least one ring of the rolling bearing. The cooling system is independent of the second lubricating circuit and is connected to the first lubricating circuit.

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Background/Summary

TECHNICAL FIELD OF THE INVENTION

(1) The present invention relates to the general field of the aeronautic. More specifically, it is aimed at an aircraft turbomachine, an equipment of which is lubricated and cooled.

TECHNICAL BACKGROUND

(2) The technical background includes in particular the documents FR-A1-3 101 684, EP-A1-2 503 107, FR-A1-3 086 326 and FR-A1-3 108 935.

(3) Conventionally, an aircraft turbomachine comprises a gas generator comprising at least one compressor, a combustion chamber and at least one turbine.

(4) The gas generator generally comprises an output shaft for driving an equipment, such as a transmission box.

(5) In this application, “equipment” means any equipment that can be coupled to the output shaft of the gas generator and comprises at least one rotor which is rotationally guided by at least one rolling bearing. A “rolling bearing” is a bearing comprising rolling elements, such as balls or rollers, typically mounted between two rings, one internal and one external.

(6) The gas generator comprises a lubricating circuit which generally comprises an oil tank, a pump and oil pipes or nozzles. The lubricating circuit of the gas generator is used, for example, to lubricate the rolling bearings of the gas generator. It is also necessary to lubricate the guiding rolling bearing of the rotor on the equipment.

(7) For regulatory reasons, the lubricating circuit of the gas generator cannot always be used to lubricate the equipment, which must therefore be equipped with an independent lubricating circuit. This is the case, for example, with a turbomachine for helicopters. The equipment then comprises its own lubricating circuit, which is used, for example, to lubricate the guide bearings of its rotor as well as any gear toothings.

(8) The circulation of lubricating oil enables the heat generated by the bearings and any lubricated gears to be extracted and cooled. If the equipment malfunctions or lubrication is interrupted, the bearings are no longer cooled and can reach high temperatures that can damage them. This is all the more true when the rotor of the equipment rotates at high speeds, typically 25,000 rpm or more.

(9) It is therefore necessary to provide a system for cooling the equipments bearings, which can ensure cooling even in the event of a malfunction in the equipments lubricating circuit.

(10) In the current state of the art, it has already been proposed to provide oil reservoirs for lubricating the bearings during flight by means of nozzles or fog, particularly in the event of an oil shortage. However, the quantities of oil available are very small and are not sufficient to lubricate and cool the bearings, particularly those guiding the rotors at high speeds.

(11) Emergency lubrication systems (using glycol-based oil) can be used but have the disadvantage of being heavy and cumbersome. In addition, this oil ensures the lubrication of the contacts but is still very much insufficient to cool high-speed rotors, which can reach high temperatures in case the lubrication is interrupted.

(12) For example, on the high-speed lines of a turbomachine, a long oil interruption means that the turbomachine has to be shut down quickly to prevent a rapid failure. However, it is not always technically possible to shut down certain equipment connected to the gas generator in flight.

(13) It has already been proposed to fit the equipment with a cooling system for its bearings. The cooling system is configured to cool the external ring of the rolling bearing only. This cooling system is currently connected to the equipment lubrication circuit, as described in the aforementioned document FR-A1-3 101 684. The cooling system allows continuous cooling of the bearing, but does not provide cooling in the event of a malfunction in the equipments lubricating circuit. The operation of the cooling system is therefore dependent on the operation of the lubricating circuit, which is problematic and does not allow safe cooling of the bearing to prevent it from overheating. This solution can also be used to cool low and medium-speed rotor guide bearings (from 400 to 20,000 rpm), but is not suitable for cooling high-speed rotors.

(14) There is therefore a need to find a solution to ensure the cooling of the rolling bearing(s) of a rotor of the equipment, even during a malfunction of the equipments lubricating circuit.

SUMMARY OF THE INVENTION

(15) The present invention proposes an aircraft turbomachine, this turbomachine comprising: a gas generator comprising at least one compressor, a combustion chamber and at least one turbine, the gas generator comprising an output shaft and a first lubricating circuit, an equipment coupled to the output shaft of the gas generator and comprising a rotor which is rotationally guided by at least one rolling bearing, the equipment comprising a second lubricating circuit which is independent of said first lubricating circuit and which is configured to lubricate said rolling bearing, the equipment further comprising a system for cooling said rolling bearing, the cooling system being configured to ensure a circulation of oil in the region of at least one ring of said rolling bearing, characterised in that the cooling system is independent of said second lubricating circuit and is connected to said first lubricating circuit.

(16) In the present application, the independent circuits are defined as circuits that do not have direct fluid communication between them, i.e. the circuits are not connected to each other. There is therefore no common valve or duct between these circuits. Preferably, the circuits do not even share the same fluid tank.

(17) Unlike the previous technique, in which the cooling system of a bearing of the equipment is independent of the lubricating circuit of the gas generator and is connected to the lubricating circuit of the equipment, the cooling system here is independent of the lubricating circuit of the equipment and is connected to the lubricating circuit of the gas generator. It is therefore understood that the bearing of the equipment can be cooled as long as the lubricating circuit of the gas generator is operational. The operation of the cooling of the bearing of the equipment is therefore independent of the operation of the lubricating circuit of the equipment. Even if the lubricating circuit of the equipment malfunctions, the cooling system of the bearing of the equipment can continue to be supplied with oil and cool the bearing of the equipment.

(18) The cooling system is configured to ensure the cooling of the guide bearing(s) of the rotors, even when the latter is rotating at high speeds of 25,000 rpm or more. The configuration of the cooling system is not constrained by the capacities of the second lubricating circuit and benefits from the capacities of the first lubricating circuit, the capacities of the first circuit generally being greater than those of the second circuit. Capacities are defined in particular as the flow rate and volume of oil available for cooling.

(19) The equipment within the meaning of the invention is preferably selected from an electrical machine, a gear or speed box, a speed reduction gear and a transmission box. For example, it can be a speed reduction gear of the RGB type (Reduction GearBox), a transmission box of the BTP or BTI type (Boîte de Transmission Principale or Intermédiaire), or a propulsion speed box of the PGB type (Propeller GearBox).

(20) The turbomachine may comprise one or more of the following characteristics, taken alone or in combination with each other. the cooling system is configured to ensure a circulation of oil in the region of an external ring of said rolling bearing; the cooling system is a permanent cooling system and is configured to ensure a continuous circulation of oil in the region of the ring(s) of said rolling bearing; the cooling system is a safety cooling system and is configured to be activated to ensure oil circulation in the region of the ring(s) of said rolling bearing; the cooling system is connected to an activation system selected from an electrical system, a thermostatic system and a hot-melt system; the cooling system comprises pipes formed in an external casing of the equipment; the pipes comprise at least one cooling pipe extending around the ring(s) of the rolling bearing, said at least one cooling pipe being connected both to at least one feed oil pipe from the first lubricating circuit and to at least one oil return pipe to the first lubricating circuit; the external ring of said rolling bearing is integrated into said casing or is mounted in said casing; said casing is fixed to the gas generator, the equipment is chosen from an electrical machine, a gear or speed box, a speed

reduction gear and a transmission box; said or each ring of the bearing is made of a metal alloy or ceramic; each of said first and second lubricating circuits comprises a dedicated oil tank and/or a dedicated pump; the oil tank of the second lubricating circuit is a gravity tank; the pump of the second lubricating circuit is configured to supply oil nozzles and to fill the gravity tank.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) Further characteristics and advantages of the invention will become apparent from the following detailed description, for the understanding of which reference is made to the attached drawings in which:

(2) FIG. **1** is a schematic axial sectional view of an aircraft turbomachine, and in particular of a gas generator and an equipment of this turbomachine;

(3) FIG. **2** is a larger-scale view of part of FIG. **1** and shows a rotor of the equipment guided by rolling bearings;

(4) FIG. **3** is a perspective view of part of an equipment intended to be coupled to an output shaft of a gas generator, and shows a system for cooling the guide bearings of its rotor,

(5) FIG. **4** is a schematic view of the equipment part in perspective and axial section of the FIG. **3**;

(6) FIG. **5** is a partial schematic view of an axial cross-section of an aircraft turbomachine, similar to FIG. **1**, and shows an alternative embodiment of the invention.

DETAILED DESCRIPTION OF THE INVENTION

(7) FIG. **1** shows a turbomachine **10** for an aircraft, in particular a helicopter. The turbomachine **10** essentially comprises two parts or modules, namely a gas generator **12**, on the right of the drawing, and an equipment **14**, on the left of the drawing.

(8) The gas generator **12** conventionally comprises, from upstream to downstream (from left to right in the drawing), with reference to the flow of gases, one or more compressors **16**, a combustion chamber **18** and one or more turbines **20**.

(9) The gas generator **12** comprises an air inlet **21**, in this case annular, which supplies air to the compressors **16**. In the example shown, the gas generator **12** comprises an axial compressor **16***a* with two compression stages followed by a centrifugal compressor **16***b*. The air is compressed in the compressors **16**, mixed with fuel and burnt in the combustion chamber **18**. The combustion gases leaving the chamber **18** feed a first turbine **20***a*, the rotor of which is connected by a common shaft **22** to the rotor of the compressors **16**. The combustion gases are expanded in the first turbine **20***a* and then in a free turbine **20***b*, the shaft **24** of which extends coaxially inside the shaft **22**, as far as the upstream end of the gas generator **10**. The shaft **24** comprises an upstream end which is located outside the gas generator **10** and forms an output shaft **26** intended to be coupled to the equipment **14**.

(10) The gas generator **12** comprises a lubricating circuit **28**, in particular for the rolling bearings (not shown) guiding the shafts **24** and **22**. This lubricating circuit **28** comprises, for example, a tank, a pump **30** and oil nozzles on the rolling bearings. The pump **30** is, for example, a pump comprising a rotor connected by a gear **32** to a power sampling device **34** on the shaft **22**.

(11) The lubricating circuit **28** of the gas generator **12** is located in the gas generator **12** and in particular inside an external casing **36** of the gas generator.

(12) The equipment **14** is preferably selected from an electrical machine, a gear or speed box, a speed reduction gear and a transmission box. For example, it can be a speed reduction gear of the RGB type (Reduction GearBox), a transmission box of the BTP or BTI type (Main or Intermediate GearBox), or a propulsion speed box of the PGB type (Propeller GearBox).

(13) The equipment **14** comprises a rotor **38** which is coupled, for example by splines, to the output shaft **26** of the gas generator **12**. The rotor **38** is rotationally guided by one or more rolling bearings

40, which can be seen more clearly in FIG. **2**. The rotor **38** comprises, for example, external helical toothing **42** located between two external cylindrical surfaces **44**. The bearings **40** are located respectively on the two surfaces **44** and comprise internal rings **40***a* mounted on these surfaces **44** and secured in rotation with the rotor **38**. The bearings **40** also comprise external rings **40***b* supported by a stator and rolling elements **40***c* arranged between the internal and external rings. The rings **40***a*, **40***b* of the bearings **40** are typically made of a metal alloy or ceramic. The equipment **14** comprises a lubricating circuit **46**, in particular for rolling bearings **40**. This lubricating circuit **46** comprises, for example, a tank, a pump **48** and oil nozzles on the rolling bearings **40**. The pump **48** is, for example, a pump comprising a rotor coupled to one end of the rotor **38**.
(14) The turbomachine **10** also comprises a system **50** for cooling the rolling bearings **40** and in particular their external rings **40***b*. This cooling system **50** is independent of the lubricating circuit **46**, i.e. it is not connected to this circuit **46**, and is instead connected to the lubricating circuit **28** of the gas generator **12**.
(15) The cooling system **50** is configured to ensure an oil circulation at the external rings **40***b* of the rolling bearings **40**.
(16) FIGS. **1** and **2** show very schematically the pipes **52** (in the form of dotted lines) which supply oil from the lubrication circuit **28** to the bearings **40**, as well as discharging this oil which can be re-injected into the lubricating circuit **28**.
(17) FIGS. **3** and **4** show a more concrete embodiment of the invention. The rotor **38** of the equipment **14** comprises external splines **54** at one end for coupling to the output shaft **26** (which comprises complementary internal splines). The rotor **38** also comprises the toothing **42** and the surfaces **44** on which the internal rings **40***a* of the bearings **40** are mounted. FIG. **4** shows that one of the bearings **40**, on the left of the drawing, is ball bearing and its external ring **40***b* is integrated into a casing **54** of the equipment **14**. The other bearing **40**, on the right of the drawing, has rollers and its external ring **40***b* is mounted in the casing **56** of the equipment **14**. The casing **56** thus forms a cage or support for the bearings **40**.
(18) The aforementioned pipes **52** of the cooling system **50** are formed in the casing **56**. This casing **56** has a general shape of revolution about the axis X of rotation of the rotor **38**. The casing **56** conforms to the general shape of the rotor **38** and comprises, for example, two adjacent cylindrical portions **56***a*, **56***b* and an external radially annular flange **56***c*.
(19) The annular flange **56***c* is located at an axial end of the casing **56**, on the side of the splines **54** of the rotor **38**, and is configured to be fixed to the casing **36** of the gas generator **12**, and for example to a corresponding annular flange of the gas generator. To achieve this, the flange **56***c* comprises an annular row of orifices for the passage of screws **58** fixed to the casing **36** of the gas generator **12**.
(20) The cylindrical portion **56***b* of larger diameter extends around the toothing **42** and the surface **44** receiving the ball bearing **40**, and from the flange **56***c* to the cylindrical portion **56***a* of smaller diameter. This portion **56***a* extends around the surface **44** receiving the ball bearing **40**.
(21) The cylindrical portion **56***b* comprises a slit **58** which exposes part of the toothing **42** and enables it to mesh with a pinion **60** or other equipment (see FIGS. **1** and **2**).
(22) Among the pipes **52** of the cooling system **50**, there are pipes **52***a* which have a generally annular shape and extend around the external rings **40***b*, when the latter are fitted in the casing **56**, or in the external rings **40***b*, when the latter are integrated in the casing **56**.
(23) The main function of the pipes **52***a* is to circulate cooling oil as close as possible to the bearings **40** in order to cool them. There may be one, two, three or more annular pipes **52***a* around each of the bearings **40**. These pipes **52***a* each have the same overall dimensions and in particular the same diameters measured in relation to the axis X.
(24) The pipes **52** also comprise feed oil pipes **52***b* formed in the casing **56**. Each of these pipes **52***b* is connected to one or more of the pipes **52***a* and comprises an end which opens at the flange **56***c* to

be fluidly connected to a corresponding end of the lubricating circuit **28** of the gas generator **12**. The lubricating circuit **28** of the gas generator **12** may also comprise an end which opens onto the flange of its casing **36** so that the assembly and fixing of the flanges ensures fluid communication between the lubrication circuit **28** and the cooling system **50**.

(25) The pipes **52** also comprise oil discharge pipes **52***c* which are formed in the casing **56**. Each of these pipes **52***c* is connected to one or more of the pipes **52***a* and comprises an end which opens at the level of the flange **56***c* to be fluidly connected to a corresponding end of the lubricating circuit **28**, in the same way as the pipes **52***b*.

(26) Alternatively, the cooling system **50** could be configured to cool the internal rings **40***a* of the bearings **40** only, or the external **40***b* and internal rings **40***a* of the bearings **40**.

(27) Generally speaking, the pipes **52** of the cooling system **50** can have any general shape and for example a serpentine or zigzag shape, and with any cross-sectional shape and for example round, oval, etc.

(28) The casing **56** shown in the drawings can be produced by additive manufacturing, although this is not restrictive.

(29) In one embodiment, the cooling system **50** can be configured to ensure continuous oil circulation at the level of the rings **40***b* of the bearings **40**.

(30) Alternatively, the cooling system **50** is a safety cooling system and is configured to be activated to ensure oil circulation at the level of the rings **40***b* of the bearings **40**. The cooling system **50** then has two distinct states: a first deactivated state in which there is no circulation of oil in the pipes **52** and therefore no cooling of the bearings **40** by heat exchange with the oil, and a second activated state in which the oil circulates in the pipes **52** to cool the bearings **40**.

(31) The cooling system **50** is then connected to an activation system selected from an electrical system, a thermostatic system and a hot-melt system. This activation system comprises, for example, one or more solenoid valves placed in the circuit and controlled by an electrical, thermostatic or hot-melt element. The solenoid valve can be designed to be activated manually or automatically. A pilot of the aircraft can, for example, control the system to be activated when a light in the cockpit signals an anomaly, this light relating, for example, to the oil pressure in the lubricating circuit **46** of the equipment **14**. The activation system can be associated with a temperature sensor to monitor the temperature of the bearings and activate the cooling system when this temperature exceeds a predetermined threshold.

(32) FIG. **5** shows another embodiment of the invention in which the lubricating circuit **46** of the equipment **14** comprises a gravity tank **62**, i.e. a tank from which oil flows by gravity to the bearings **40** in particular. Once this oil has lubricated the bearings, it is recovered and can be returned to the gravity tank **62** using the pump **48** in the circuit **46** (see arrow F). This pump **48** can also be used to supply oil nozzles to the bearings **40** and to other elements to be lubricated on the equipment **14**. The oil supplied by the gravity tank **62** can then be used to lubricate the bearings **40** in parallel, in addition to the main lubrication provided by the oil nozzles (connected to the tank **62** or to another tank in the circuit **46**). The embodiment shown in FIG. **5** also comprises a cooling system **50** as described above.

(33) As can be seen in the example shown, the gravity tank **62** can be mounted in the gas generator **12** and in particular in its casing **36**. It is then connected to bearing lubrication means by one or more pipes **64**, which comprise a top end connected to the tank **62** and a bottom end connected to these lubrication means for the gravitational flow of oil from the tank **62** to the bearings **40**.

(34) The invention thus proposes to convey lubricating oil from the gas generator **12** into the equipment **14** to cool the bearings **40** of this equipment. The heat generated by the bearings is then removed by the lubricating oil, which is then cooled in the gas generator **12**.

Claims

1. An aircraft turbomachine, comprising: a gas generator comprising at least one compressor, a combustion chamber, and at least one turbine, the gas generator comprising an output shaft and a first lubricating circuit, and an equipment coupled to the output shaft of the gas generator and comprising a rotor which is rotationally guided by at least one rolling bearing, the equipment comprising a second lubricating circuit which is independent of said first lubricating circuit and which is configured to lubricate said at least one rolling bearing, the equipment further comprising a system for cooling configured to cool said at least one rolling bearing, the cooling system being configured to ensure a circulation of oil in the region of at least one ring of said at least one rolling bearing, wherein the cooling system is independent of said second lubricating circuit and is connected to said first lubricating circuit.
2. The turbomachine according to claim 1, wherein the cooling system is configured to ensure a circulation of oil in the region of an external ring of said at least one rolling bearing.
3. The turbomachine according to claim 1, wherein the cooling system is a permanent cooling system and is configured to ensure a continuous circulation of oil in the region of the ring(s) of said at least one rolling bearing.
4. The turbomachine according to claim 1, wherein the cooling system is a safety cooling system and is configured to be activated to ensure a circulation of oil in the region of the ring(s) of said at least one rolling bearing.
5. The turbomachine according to claim 4, wherein the cooling system is connected to an activation system which is chosen from an electrical system, a thermostatic system, and a hot-melt system.
6. The turbomachine according to claim 1, wherein the cooling system comprises pipes formed in an external casing of the equipment.
7. The turbomachine according to claim 6, wherein the pipes comprise at least one cooling pipe extending around the ring(s) of the at least one rolling bearing, said at least one cooling pipe being connected both to at least one feed oil pipe from the first lubricating circuit, and to at least one oil return pipe to the first lubricating circuit.
8. The turbomachine according claim 2, wherein the cooling system comprises pipes formed in an external casing of the equipment, and the external ring of said at least one rolling bearing is integrated into said casing or is mounted in said casing.
9. The turbomachine according to claim 6, wherein said casing is fixed to the gas generator.
10. The turbomachine according to claim 1, wherein the equipment is chosen from an electrical machine, a gear or speed box, a speed reduction gear, and a transmission box.
11. The turbomachine according to claim 1, wherein said or each ring of said at least one rolling bearing is made of a metal alloy or ceramic.
12. The turbomachine according to claim 1, wherein each of said first and second lubricating circuits comprises a dedicated oil tank.
13. The turbomachine according to claim 12, wherein the oil tank of the second lubricating circuit is a gravity tank.
14. The turbomachine according to claim 1, wherein each of said first and second lubricating circuits comprises a dedicated pump.
15. The turbomachine according claim 13, wherein each of said first and second lubricating circuits comprises a dedicated pump, and the pump of the second lubricating circuit is configured to supply oil nozzles and to fill the gravity tank.